

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

Fourth Semester of B. Tech. Examination (CL)

December 2011

CL209 Structural Analysis-I

Date: 07.12.2011, Wednesday

Time: 10:00 a.m. To 01:00 p.m.

Maximum Marks: 70

Instructions:

1. The question paper comprises of two sections.
2. Section I and II must be attempted in separate answer sheets.
3. Make suitable assumptions and draw neat figures wherever required.
4. Use of scientific calculator is allowed.

SECTION - I

Q - 1 Differentiate between determinate and indeterminate structure. Give advantages and disadvantages of indeterminate structures. [07]

Q - 2 (a) Solve any two [08]

1. A thin cylindrical cell with internal diameter 125mm and wall thickness 12mm is subjected to a steam pressure 10N/mm^2 . Find the circumferential and longitudinal stress in the shell material.

2. A water main of 800mm internal diameter is carrying water at pressure head of 100mm. If the allowable hoop stress is 20 N/mm^2 , find suitable thickness of the pipe. Take density of water as 10 kN/m^3 .

3. A thin cylindrical shell 3000mm long 1200mm in diameter is subjected to an internal pressure of 1.5 N/mm^2 . If the thickness of the shell is 12mm, find the circumferential and longitudinal strains developed and hence find the increase in the capacity of shell.

(b) A rectangular column of size 500mm x 250mm carries an eccentric load of 1000kN on the axis bisecting the thickness at 150mm from centroidal axis. Find the maximum and minimum resultant stress. [06]

OR

(b) A cast iron column having 150mm diameter carries an eccentric load of 50kN. If maximum tensile stress is not to exceed 7.5 N/mm^2 , find the permissible eccentricity of load on column. [06]

Q - 3 (a) A circular column of diameter d is subjected to a load of 200kN at an eccentricity of $0.3d$. If the maximum stress at the base of the column is limited to 50N/mm^2 , find out the diameter of column. [04]

(b) Define Column, Strut, Radius of Gyration and Slenderness ratio. [04]

(c) A rectangular column 400mm wide and 300mm deep is fixed at both the ends. The length of column is 6m. If $E = 1.2 \times 10^5\text{ N/mm}^2$. Find Euler's Crippling load. [06]

OR

(c) A hollow circular section is having internal diameter 60mm and 10mm thickness. Calculate radius of gyration. [06]

SECTION - II

- Q - 4** A beam simply supported over a span of 6m is carrying a point load of 50kN at 1.2m from right support. Find the position and amount of maximum deflection. Also calculate deflection at mid span. Take $EI = 20,000 \text{ kN.m}^2$. [07]
- Q - 5 (a)** Derive relation between slope, deflection and radius of curvature. [05]
- (b)** Draw influence line diagram of R_a , R_b , V_x and M_x at section 3m from left support of a simply supported beam AB of length 8m. [08]

OR

- (b)** Draw influence line diagram of R_a , R_b , V_x and M_x at section 2m from left support of a simply supported beam AB of length 5m. Also find the values of R_a , R_b , V_x and M_x when a series of four concentrated loads 10kN, 12kN, 15kN and 8kN are acting at 1m, 2m, 3m and 4m from A. [08]
- Q - 6 (a)** Explain how the udl should be placed for obtaining maximum bending moment at a section. [04]
- (b)** A three hinge parabolic arch has span 20m and central rise 3m. It carries a point load of 10kN at 7.5m from the left hinge. Calculate the normal thrust, shear and bending moment at section 7.5m from right hand hinge. Also calculate maximum positive and negative bending moments and their positions. [11]
